Kernel Geometry Governs Robustness under Distribution Shift: A Controlled Multi-Dataset Study of Quantum and Classical Kernels

Roberto Fernández-Barrios[ORCID][1*], Iker Pastor-López[ORCID][1], Asier González-Santocildes[ORCID][1], Pablo García Bringas[ORCID][1]

[1*]Faculty of Engineering, University of Deusto, Avda. de las Universidades, 24., Bilbao, 48007, Spain.

*Corresponding author(s). E-mail(s): roberto.fernandez.b@deusto.es; Contributing authors: iker.pastor@deusto.es; gonzalez.asier@deusto.es; pablo.garcia.bringas@deusto.es;

Abstract

Whether quantum kernels improve robustness under distribution shift is usually asked as a quantum-versus-classical question. We argue, and show empirically, that it is more productively asked as a question about kernel geometry. We present a controlled kernel-swap study in which the classifier, preprocessing, split construction, and family-internal model-selection logic are held fixed while only the kernel varies, instantiated on four security benchmarks spanning two modalities and three drift mechanisms (EMBER static malware; UNSW-NB15 and ToN-IoT network flows under synthetic covariate shift and natural regime drift), two classifier families (support vector classification and a Laplace-approximation Gaussian process classifier with calibrated uncertainty), and 72 principal settings with 15 repeated runs each; all quantum kernels are computed by exact statevector simulation, and no hardware claim is made. Against the customary linear and RBF baselines, fidelity-based quantum kernels improve out-of-distribution balanced accuracy nearly uniformly (69–71 of 72 settings; Wilcoxon $\boldsymbol{p < 10^{-9}}$ on network data). Against a broader classical family, a median-heuristic Laplacian kernel neutralizes the advantage on EMBER—revealing that the operative property is not quantumness but geometry—yet when both families receive the same bandwidth-tuning freedom the quantum advantage re-emerges on

EMBER and remains dominant under natural network drift (54 of 54 settings under the probabilistic classifier, $\boldsymbol{p = 1.6 \times 10^{-10}}$). A mechanistic analysis explains these outcomes: within each setting, out-of-distribution accuracy tracks the survival of centered kernel-target alignment under shift, a regularity that holds in 89–99% of 360 setting-seed units across all four datasets, with high effective rank of the training kernel as its structural precondition. The classical-favorable exceptions of the original comparison are precisely the settings where the selected quantum kernel collapses to low-rank, classical-like geometry. Beyond the specific quantum-versus-classical verdict, the study contributes a reusable, fully reproducible protocol for attributing robustness effects to kernel geometry, and a design principle—prefer kernels whose label-aligned geometry survives shift—that spans classical and quantum families alike.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods are one of the clearest and most actively studied directions in quantum machine learning. By embedding classical data into quantum states and defining similarity through state overlaps, they provide a principled way to study how quantum-induced feature geometry affects supervised learning while keeping the downstream classifier fixed Havlíček et al. (2019); Schuld and Killoran (2019); Schuld (2021). At the same time, the recent QML literature has made an equally important point: empirical conclusions about quantum models can depend strongly on the data regime, the encoding, and the evaluation protocol Huang et al. (2021); Schuld et al. (2021); Gil-Fuster et al. (2024); Bowles et al. (2024); Schnabel and Roth (2025). For that reason, claims about the usefulness of quantum kernels are most informative when they are established under controlled, reproducible, and carefully matched comparisons rather than through heterogeneous pipelines or loosely aligned baselines.

This need for control becomes even more important under distribution shift. In many applied settings, the relevant question is not only whether a model separates classes well under in-distribution (ID) conditions, but whether its behavior remains reliable once the evaluation distribution changes. The broader distribution-shift and domain-generalization literature has repeatedly shown that apparent robustness gains can weaken, disappear, or become difficult to interpret when model-selection strategy, hyperparameter search, split construction, or evaluation standardization are not carefully controlled Quiñonero-Candela et al. (2008); Zhou et al. (2023); Gulrajani and Lopez-Paz (2021); Koh et al. (2021); Yao et al. (2022). A convincing robustness study must therefore satisfy two requirements at once: it must isolate the factor of interest tightly enough to support attribution, and it must evaluate that factor broadly enough to determine whether the observed behavior reflects a meaningful regime-level effect rather than a narrow configuration artifact.

These issues are especially acute in security applications. Drift is not incidental to the problem; it is part of the problem definition itself. Static PE malware classification has been widely studied through benchmarks such as EMBER Anderson and Roth (2018), while more recent work has highlighted the gap between laboratory evaluation and performance on drifted malware, as well as the difficulty of adapting with limited post-drift data Barbero et al. (2022); Li et al. (2025). The emergence of updated resources such as EMBER2024 further reflects an active push toward more realistic, broader, and more deployment-relevant malware evaluation Joyce et al. (2025). Network intrusion detection faces the same structural challenge: traffic distributions move as services, infrastructure, and attack campaigns evolve, and public benchmarks such as UNSW-NB15 and ToN-IoT expose exactly this kind of regime change Moustafa and Slay (2015); Alsaedi et al. (2020). Taken together, this literature suggests that a useful security-ML study should not merely report accuracy on a convenient split; it should examine how model choices behave when the evaluation explicitly targets shift, and it should do so across more than one data modality and drift mechanism.

Kernel methods are a natural setting in which to ask such questions cleanly. Because kernel-based support vector machines separate the choice of classifier from the choice of similarity function, they offer a principled way to study how representational geometry affects ID and OOD behavior while keeping the learning machinery fixed Cortes and Vapnik (1995); Schölkopf and Smola (2002). This perspective becomes particularly interesting in quantum machine learning, where quantum kernels are defined through implicit feature maps into quantum Hilbert spaces and estimated via state overlaps or related fidelity-based similarities Havlíček et al. (2019); Schuld and Killoran (2019); Schuld (2021). From this viewpoint, quantum kernels are not a vague alternative model class, but a concrete family of kernel constructions whose empirical value can—and should—be tested under the same standards applied to classical kernels.

At the same time, the quantum-kernel literature has matured to the point where optimistic demonstrations and critical scrutiny must now be read together. Foundational work established the feature-space interpretation of quantum kernels and motivated them as expressive alternatives to classical similarities Havlíček et al. (2019); Schuld and Killoran (2019). Subsequent studies emphasized that performance in quantum machine learning can depend strongly on the data regime, the encoding, and the effective geometry induced by the model Huang et al. (2021); Schuld et al. (2021). More recent work has sharpened this message further: understanding generalization in quantum machine learning requires care beyond classical intuition, and benchmarking conclusions can be highly sensitive to protocol choices Gil-Fuster et al. (2024); Bowles et al. (2024); Schnabel and Roth (2025). This makes broad claims of superiority difficult to defend and reinforces the need for conservative, protocol-aware empirical studies.

The gap we address lies precisely at this intersection, and it has three parts. First, there remains limited evidence from controlled, variability-aware comparisons between classical and quantum kernel families in security-relevant settings where out-of-distribution robustness is part of the evaluation target: existing studies often differ simultaneously in model class, preprocessing, search space, or data protocol, making it difficult to attribute observed gains to the kernel itself. Second, when such comparisons are attempted, the classical reference set is typically narrow—linear and RBF kernels under a single classifier—so an apparent quantum advantage may reflect the weakness of the baselines rather than a property of the quantum family; probabilistic classifiers with principled uncertainty quantification, and classical kernels with richer spectral structure, are usually absent. Third, and most fundamentally, empirical wins and losses are rarely accompanied by an account of *why* a kernel family should behave better or worse under shift, leaving results as observations in search of an explanation. For a question as specific as whether quantum kernels help under distribution shift, these confounds and omissions are not peripheral; they are decisive.

In this paper, we study that question through a strictly controlled *kernel-swap* protocol, instantiated broadly enough to close all three gaps at once. Within each experimental setting we hold the classifier, the preprocessing pipeline, the split definitions, and the family-internal model-selection logic fixed, and vary only the kernel. The protocol is applied to four security benchmarks spanning two data modalities and three drift mechanisms: EMBER static PE malware under two synthetic covariate-shift constructions Anderson and Roth (2018), and UNSW-NB15 and ToN-IoT network-flow scenarios under both an analogous synthetic construction and *natural* regime drift, in which the out-of-distribution pool consists of genuinely drifted traffic Moustafa and Slay (2015); Alsaedi et al. (2020). Two classifier families consume the same precomputed kernels: support vector classification, and a binary Laplace-approximation Gaussian process classifier Rasmussen and Williams (2006) that adds calibrated predictive probabilities—and hence principled uncertainty quantification under shift—without changing anything else in the pipeline. The classical family ranges from the customary linear and RBF baselines to polynomial, Laplacian, and Matérn kernels with train-only median-heuristic length scales; the quantum family comprises fidelity kernels induced by commonly used feature-map families. Crucially, both families receive the same bandwidth-tuning freedom (a gamma sweep for RBF, an angle-scaling sweep for the quantum maps), so that neither side is compared at a fixed operating point the other is allowed to tune away from Shaydulin and Wild (2022); Canatar et al. (2023). The main evaluation comprises 72 principal settings (18 on EMBER, 54 on network flows), each repeated across 15 runs, with paired Wilcoxon signed-rank tests and Holm correction on all family-level comparisons.

Beyond measuring who wins, we ask why. For every kernel, dimension, and setting we compute geometry descriptors of the induced Gram matrices—the effective rank of the training kernel Roy and Vetterli (2007), centered kernel-target alignment on the ID and OOD splits Cristianini et al. (2001); Cortes et al. (2012), kernel concentration statistics, and the geometric difference between classical and quantum kernels Huang et al. (2021)—and test which of these quantities predict out-of-distribution behavior within and across settings. This mechanistic layer converts the benchmark from a scoreboard into an instrument for attribution.

The resulting picture is sharper than a quantum-versus-classical verdict, and more useful. Against linear and RBF baselines, fidelity-based quantum kernels improve OOD balanced accuracy nearly uniformly across all four datasets (69–71 of 72 settings under both classifiers). Against the extended classical family, the advantage on EMBER is neutralized by a median-heuristic Laplacian kernel—demonstrating that the operative property is geometric rather than quantum—but re-emerges under symmetric bandwidth tuning, and remains dominant on network flows, where under the probabilistic classifier the quantum family wins all 54 settings with gains exceeding run-to-run variability in the majority. In-distribution separability favors the quantum family in every EMBER setting regardless of the classical reference set. The mechanistic analysis explains all of these outcomes with one regularity: within a setting, OOD accuracy tracks the survival of centered kernel-target alignment under shift—positive in 89–99%

of 360 setting-seed units across all four datasets and both classifiers—with high effective rank of the training kernel as its structural precondition. The classical-favorable exceptions of the baseline comparison are exactly the settings where the selected quantum kernel collapses to a low-rank, classical-like geometry, and the Laplacian is exactly the classical kernel whose geometry most resembles the quantum maps.

Our contributions are fourfold. First, a controlled, reproducible kernel-swap protocol for robustness studies that extends across classifier families (including a probabilistic classifier with calibrated uncertainty), across a broad classical reference set with symmetric bandwidth tuning, and across datasets, modalities, and drift mechanisms. Second, a variability-aware regime map of quantum-versus-classical kernel behavior under shift on 72 settings with formal significance testing, including the honest negative result that a well-chosen classical kernel matches the quantum family on one modality at fixed bandwidth. Third, a mechanistic account—alignment survival under shift, enabled by high effective rank—that predicts within-setting OOD performance across every dataset we study, explains the structured exceptions, and yields the checkable prediction (which we confirm) that the classical kernels sharing this geometry are the competitive ones. Fourth, practical guidance and a fully public artifact: for OOD-sensitive kernel selection, prefer kernels whose label-aligned geometry survives shift; fidelity-based quantum maps exhibit this property natively, and on drifting network traffic they currently exploit it best.

2 Related Work

2.1 Distribution shift, domain generalization, and evaluation rigor

The broader literature on distribution shift has established that predictive performance can degrade substantially when training and test data are drawn from different distributions, and that robust evaluation under shift requires more than reporting strong in-distribution performance Quiñonero-Candela et al. (2008); Shimodaira (2000); Sugiyama et al. (2007). Within machine learning, domain generalization (DG) has emerged as one of the main formal settings for studying out-of-distribution (OOD) performance without access to target-domain labels during training Zhou et al. (2023). This literature has also made clear that empirical conclusions about robustness are highly sensitive to evaluation design.

A particularly important development is the move toward benchmark-driven scrutiny of robustness claims: DomainBed showed that many reported domain-generalization improvements dissolve once datasets, architectures, hyperparameter budgets, and model-selection criteria are held consistent Gulrajani and Lopez-Paz (2021); WILDS demonstrated that standard training leaves large OOD gaps under naturally occurring real-world shifts Koh et al. (2021); and Wild-Time established temporal shift as a distinct, practically relevant regime that existing methods struggle to close Yao et al. (2022). These works matter here not because security detection is a standard DG benchmark, but because they establish a methodological principle we adopt wholesale: when the goal is to compare robustness under shift, fairness of protocol is part of the contribution. Our comparison therefore fixes the classifier family, preprocessing, and split definitions, so that the role of the kernel can be interpreted with minimal confounding.

2.2 Security detection under drift and evolving evaluation benchmarks

In malware detection, the relevance of distribution shift is especially acute. Malware ecosystems evolve continuously as attackers modify payloads, packers, obfuscation strategies, delivery mechanisms, and development toolchains. Static detection models are therefore exposed to changing distributions in ways that are operationally consequential rather than merely academic. Public benchmarks have played an important role in making this problem more reproducible. EMBER remains one of the most widely used open datasets for static PE malware detection and helped standardize large-scale evaluation in this area Anderson and Roth (2018). More recent resources such as EMBER2024 push this effort further by broadening scope, increasing scale, and enabling more holistic evaluation across multiple malware-related tasks Joyce et al. (2025). Other datasets such as BODMAS and SOREL-20M likewise reflect the field's effort to move toward larger and more realistic benchmark settings Yang et al. (2021); Harang and Rudd (2020).

At the same time, a growing body of work has shown that strong laboratory performance does not guarantee robustness to evolving malware. Barbero et al. revisited malware classification under concept drift and showed that deployment-oriented settings raise challenges that are obscured by simpler static evaluations Barbero et al. (2022). More recently, Li et al. studied adaptation to real drifted malware under limited post-drift supervision, reinforcing the point that temporal degradation is not hypothetical and that realistic recovery from drift may be data-constrained Li et al. (2025). Taken together, these studies support a clear conclusion: in malware detection, robustness under shift is not a niche desideratum but a primary evaluation target.

The same structural problem, with faster drift dynamics, defines network intrusion detection: benchmark corpora such as UNSW-NB15 and ToN-IoT capture traffic whose attack composition and background behavior change across collection regimes Moustafa and Slay (2015); Alsaedi et al. (2020), and adaptive NIDS pipelines are routinely retrained for exactly this reason. Using both modalities in one protocol lets us ask whether kernel-level robustness effects are properties of a dataset family or of the kernels.

Our work differs from this line in emphasis. Rather than proposing a new drift-adaptation mechanism, drift detector, or continual-update strategy, we ask a narrower but complementary question: under a tightly controlled protocol, does the choice of kernel family itself materially affect OOD robustness? This question is well motivated by the security literature, yet comparatively underexplored in a way that isolates kernels from the many other moving parts of the pipeline.

2.3 Kernel methods, quantum kernels, and careful benchmarking in QML

Kernel methods provide a natural framework for studying robustness under controlled model comparisons because they separate the learning algorithm from the similarity function used to compare inputs. In classical machine learning, support vector machines and related kernel methods have long provided a principled route for controlling model capacity while varying the geometry of the induced feature space Cortes and Vapnik (1995); Schölkopf and Smola (2002). This makes kernels especially attractive when the scientific question concerns representation and similarity rather than wholesale changes in classifier architecture.

Quantum kernel methods arise from the observation that data can be embedded into quantum states and compared through overlaps in a quantum feature space. Havlíček et al. popularized this perspective in the context of supervised classification with quantum-enhanced feature spaces Havlíček et al. (2019), while Schuld and Killoran provided a broader theoretical account linking quantum models to kernel methods through feature Hilbert spaces Schuld and Killoran (2019). Later work sharpened this connection by arguing that supervised quantum machine learning models can often be interpreted directly as kernel methods, making the kernel viewpoint central rather than incidental Schuld (2021). Additional work on data encoding further showed that representational behavior depends critically on how classical inputs are mapped into quantum states, which is directly relevant when comparing quantum feature maps as alternative kernel constructions Schuld et al. (2021).

However, the contemporary QML literature also makes clear that promising empirical behavior should not be read uncritically. Huang et al. showed that the role of data can be decisive in quantum machine learning, suggesting that observed performance gains are deeply regime-dependent rather than model-intrinsic Huang et al. (2021). Caro et al. studied generalization from few training examples, while Gil-Fuster et al. later argued that understanding generalization in QML requires rethinking familiar classical intuitions, since standard explanations may fail to account for observed behavior in quantum models Caro et al. (2022); Gil-Fuster et al. (2024). In parallel, broader perspective pieces have emphasized both the promise and the unresolved challenges of QML as a field Cerezo et al. (2022).

Most importantly for the present work, recent benchmarking studies have raised the standard for empirical claims in QML. Bowles et al. argued that experimental design choices can dominate conclusions and showed, in a broad simulated benchmark, that out-of-the-box classical models often remain strong baselines against quantum alternatives Bowles et al. (2024). Schnabel and Roth extended this benchmarking perspective specifically to quantum kernel methods, conducting a large-scale study across many design choices and showing that performance depends strongly on kernel type, encoding strategy, and hyperparameter configuration rather than on a simple classical-versus-quantum dichotomy Schnabel and Roth (2025). Related controlled evidence from adjacent QML settings points in the same methodological direction: recent work in hybrid quantum reinforcement learning shows that varying multiple pipeline components at once can obscure attribution, and highlights the importance of component-isolating, reproducible evaluation under fixed experimental conditions Lazaro et al. (2025). This line of work is especially relevant to our paper because it supports a conservative framing: the value of quantum kernels, if any, must be demonstrated under controlled and transparent comparisons, and should be expected to be regime-dependent.

Three further strands supply the analytical tools of this study. First, work on kernel bandwidth in QML has shown that rescaling the data encoding is as consequential for fidelity kernels as the length scale is for classical kernels, and can determine whether quantum models generalize at all Shaydulin and Wild (2022); Canatar et al. (2023); we operationalize this insight as a fairness requirement—symmetric bandwidth freedom for both families—rather than as a quantum-side tuning trick. Second, kernel-target alignment Cristianini et al. (2001); Cortes et al. (2012) and effective rank Roy and Vetterli (2007) are classical tools for characterizing Gram-matrix geometry; to our knowledge their joint use as *predictors of OOD behavior under distribution shift*, across classical and quantum kernels alike, is new. Third, Gaussian process classification Rasmussen and Williams (2006) provides the standard route to calibrated uncertainty in kernel methods; including a Laplace-approximation GP classifier in the kernel-swap protocol lets us evaluate uncertainty quality under shift without changing anything else in the pipeline.

Our study is positioned at the intersection of these literatures. From the distribution-shift literature, we adopt the principle that robustness claims must rest on evaluation protocols that control major confounds. From the security literature, we take the practical motivation that temporal and distributional change are intrinsic to realistic deployment, in malware and network intrusion detection alike. From the QML literature, we take both the motivation for studying quantum kernels and the caution that benchmarking quality is central to interpretation. Relative to prior work, our contribution is not to claim universal superiority for quantum kernels, nor to propose a new adaptation method for drifted security data, but to provide a controlled, variability-aware, multi-dataset comparison of classical and quantum kernel families in which out-of-distribution robustness is the primary evaluation target—and to accompany the comparison with a mechanistic account of why its outcomes are what they are.

3 Methods

3.1 Problem setting

We study binary security classification—malware versus benign software, and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution data-generating process and D_{OOD} an out-of-distribution process induced by a shift mechanism, which may be synthetically constructed or naturally occurring. Given a training set $T \sim D_{\mathrm{ID}}$, we evaluate the learned classifier on both an ID test set S_{ID} and an OOD test set S_{OOD}. Our objective is not only to measure separability under the i.i.d. assumption, but to compare how kernel families behave when generalization is evaluated explicitly under shift.

The central methodological requirement of the study is attribution. Because the scientific question concerns the effect of the kernel, comparisons must avoid confounds from changes in classifier family, preprocessing, optimization protocol, or split design. We therefore adopt a strictly controlled *kernel-swap* protocol in which all components of the pipeline are held fixed except the kernel function itself. This design follows the same evaluation logic that motivates rigorous OOD benchmarking in the broader distribution-shift literature: when the target of inference is robustness, protocol control is not an implementation detail but part of the contribution.

3.2 Datasets and split construction

The protocol is instantiated on four benchmark scenarios spanning two data modalities. The first is EMBER, the standard public benchmark for static PE malware detection Anderson and Roth (2018), which is widely used, supports reproducible feature-based experimentation, and has motivated later benchmark extensions such as EMBER2024 Joyce et al. (2025). The remaining three are network-flow intrusion scenarios: two derived from UNSW-NB15 (with DoS and Reconnaissance, respectively, as the drifting attack category) Moustafa and Slay (2015) and one from ToN-IoT (Scanning) Alsaedi et al. (2020). In all cases our goal is not to benchmark absolute detection performance on a standard split, but to construct controlled ID/OOD regimes that allow kernel-family comparisons under distribution shift.

For each experimental setting, we adopt a fixed three-way split design:

$$T, \quad S_{\mathrm{ID}}, \quad S_{\mathrm{OOD}},$$

where T is used for training, S_{ID} measures in-distribution performance, and S_{OOD} measures robustness under the induced shift. All models compared within a setting use exactly the same split definitions. This point is essential: a classical and a quantum model are never compared on different train/test partitions.

EMBER shift constructions.

On EMBER we operationalize shift through two controlled OOD-construction variants, denoted $m1$ and $m2$, both built on the histogram + byte-entropy feature blocks and both designed to preserve class balance across splits.

Variant $m1$: sparsity-score extremes within class.

For each sample, we compute a sparsity-related score on a predefined feature block. In the main configuration used in the experiments, this score corresponds to the number of non-zero entries within the selected block. The OOD set is formed by taking, within each class separately, the low-score and high-score extremes, i.e., the two tails of the within-class score distribution. The remaining samples form an ID pool from which the training and ID test sets are obtained by stratified sampling. Intuitively, this construction creates an OOD regime driven by atypical sparsity structure while keeping the label distribution under control.

Variant $m2$: distance-to-train extremes within class.

We first fix the final training set at an exact target size. Using only this training set, we compute class-conditional centroids and assign each non-training sample a score equal to its Euclidean distance to the centroid of its own class. To reduce noise and avoid leakage, this distance can be computed in a TruncatedSVD-projected space fitted on the training set only. The OOD set is then defined, again within each class separately, by the low- and high-distance extremes from the non-training pool, with exact target OOD size and proportional class allocation. The ID test set is drawn from the remainder. This produces a complementary shift regime in which OOD samples are unusual relative to the final training geometry rather than to a sparsity statistic alone.

Network-flow scenarios and shift constructions.

Each network scenario is defined by a *reference* pool (traffic from the pre-drift regime) and a *current* pool (traffic containing the drifted attack campaign), prepared with a numeric-only feature policy that removes identifiers, label-derived columns, and categorical protocol fields; UNSW-NB15 scenarios retain 39 flow features and ToN-IoT 89. We verified explicitly that no feature is label-derived: the maximum feature–label correlation across scenarios is 0.62 (attained by well-known legitimate features such as `sttl`), far from the near-perfect correlation that would signal leakage, and the verification script is part of the public artifact. On these pools we instantiate two shift mechanisms. The first, *m2-centroid*, is the direct analog of EMBER's $m2$: within the class-balanced reference pool, a provisional training pool is fixed, class-conditional centroids are computed in a train-only MaxAbs+TruncatedSVD space, and the OOD pool consists of the within-class low- and high-distance tails. The second, *natural drift*, uses no synthetic construction at all: the training and ID pools are drawn from the reference regime and the OOD pool from the class-balanced current regime, so the evaluation measures robustness to genuinely drifted traffic rather than to constructed covariate extremes.

Together, these constructions are not intended as universal models of real-world drift. Rather, they provide reproducible shift mechanisms of three distinct kinds—atypical-sparsity extremes, train-geometry extremes, and natural regime drift—that are sufficiently different to support regime-level conclusions while preserving comparability across kernel families. In every setting, all models are compared on exactly the same T, S_{ID}, and S_{OOD}.

3.3 Controlled kernel-swap protocol

Our core experimental principle is to isolate the kernel as the only variable of interest. Accordingly, all compared models share the following components:

- **Classifier:** within each comparison, a single classifier consuming precomputed Gram matrices. Two classifier families are studied in parallel: Support Vector Classification with `kernel="precomputed"`, and a binary Gaussian process classifier under the Laplace approximation.
- **Preprocessing:** the same scaling and dimensionality-reduction pipeline for all classical and quantum models.
- **Split design:** identical T, S_{ID}, and S_{OOD} within each setting.
- **Selection logic:** the same family-internal model-selection principle applied consistently to classical and quantum candidates.

This yields a comparison between kernel families rather than between heterogeneous model pipelines. In particular, it prevents observed OOD differences from being attributed to changes in classifier architecture, feature engineering, or split construction. The choice of precomputed-kernel classifiers is deliberate: it allows both classical and quantum similarities to be consumed by exactly the same decision rule, thereby making the comparison interpretable in kernel-method terms Cortes and Vapnik (1995); Schölkopf and Smola (2002).

The Gaussian process classifier implements the standard binary Laplace approximation with a logistic link (Algorithms 3.1 and 3.2 in Rasmussen and Williams (2006)), applied directly to the same unit-diagonal precomputed kernels as the SVC. Its role in the study is twofold. Methodologically, it tests whether the kernel-family conclusions are classifier-specific or transfer across learning machinery. Substantively, it produces calibrated predictive probabilities, which allows robustness under shift to be examined through the lens of uncertainty quantification—log-loss, Brier score, expected calibration error, and predictive entropy on ID versus OOD data—rather than through accuracy alone. Kernel-swap comparisons are always performed within a classifier, never across classifiers.

Although the classical and quantum candidate sets are not identical in cardinality, fairness in this study is defined at the protocol level: both families are evaluated under the same classifier, preprocessing, splits, dimensionality sweep, and family-internal selection criterion, so that the effect attributed to the kernel is not confounded by changes elsewhere in the pipeline.

3.4 Preprocessing and feature representation

All models use the same preprocessing pipeline. Let $x \in \mathbb{R}^p$ denote the original feature vector of the dataset at hand. We apply the following steps:

1. **MaxAbs scaling.** We first apply `MaxAbsScaler` to normalize feature magnitudes while preserving sparsity structure.

2. **Dimensionality reduction.** We project the scaled features using `TruncatedSVD` to a reduced representation of dimension

$$d \in \{4, 6, 8, 10, 12\}.$$

3. **Angle mapping.** For the quantum pipeline, the reduced coordinates are linearly mapped to an angle range $[0, \pi]$ before being passed to the quantum feature maps.

This shared preprocessing serves two purposes. First, it keeps the comparison fair by ensuring that classical and quantum kernels operate on the same reduced representation. Second, it makes representation size itself a controlled factor of the study, enabling both aggregated setting-level comparisons and dimension-level analyses. The dimensionality parameter d is therefore not a tuning artifact hidden in preprocessing, but an explicit axis of the experimental grid.

3.5 Kernel families

Baseline classical kernels.

The customary reference set consists of two standard baselines:

$$K_{\text{lin}}(x, x') = x^\top x', \tag{1}$$

$$K_{\text{rbf}}(x, x') = \exp\left(-\gamma \|x - x'\|_2^2\right), \tag{2}$$

with the RBF kernel evaluated using the standard `scale` convention for γ and the linear kernel cosine-normalized to unit diagonal. These are the baselines against which quantum kernels are most commonly compared, and they define our *classical-original* family.

Extended classical family.

Because an apparent quantum advantage over two baselines may say more about the baselines than about the quantum family, we additionally evaluate a broader *classical-extended* family: polynomial kernels of degree 2 and 3 (cosine-normalized), the Laplacian kernel $K(x, x') = \exp(-\|x - x'\|_1/\ell_1)$, and Matérn kernels with $\nu \in \{3/2, 5/2\}$, where all length scales ℓ are set by the median heuristic—the median pairwise distance on the training set—so that every hyperparameter is determined from training data alone, without access to ID or OOD test information. These kernels were chosen deliberately to span a range of spectral behaviors: the Laplacian and low-ν Matérn kernels are less smooth than the RBF and induce Gram matrices with substantially heavier eigenvalue tails, which, as the mechanistic analysis will show, is precisely the property at stake under shift.

Bandwidth symmetry.

Kernel bandwidth is known to be a decisive hyperparameter for quantum kernels: rescaling the input encoding plays the same role for fidelity kernels that γ plays for the RBF kernel, and can qualitatively change generalization Shaydulin and Wild (2022); Canatar et al. (2023). A comparison in which the classical family may tune its bandwidth while the quantum family is pinned to a fixed encoding scale (or vice versa) is therefore not protocol-fair. In the bandwidth-symmetric portion of the study, both families receive the same tuning axis: the RBF kernel is evaluated at $\gamma = f \cdot \gamma_{\text{scale}}$ for $f \in \{0.1, 0.3, 1, 3, 10\}$, and every quantum feature map is evaluated at angle scalings $\{0.5, 1, 2\}$ of the canonical $[0, \pi]$ encoding. Family-internal selection then operates over the enlarged candidate sets exactly as before.

Quantum kernels.

Quantum kernels are defined through an implicit feature map $\phi_Q(\cdot)$ induced by a parameterized quantum circuit, with similarity measured through fidelity-like overlaps

Havlíček et al. (2019); Schuld and Killoran (2019); Schuld (2021). For two inputs x and x', the corresponding kernel entry can be written abstractly as

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2,$$

or equivalently through the overlap of states prepared by input-dependent quantum feature maps.

In the experiments, we evaluate multiple commonly used fidelity-based feature-map families chosen to represent distinct inductive biases: ZZ-type maps, PauliXZ-type maps, and Z-map variants. Concretely, the candidate set includes two ZZ configurations with different repetition depths, one PauliXZ configuration, and one Z-map configuration. We emphasize that the study is not attempting to exhaust the quantum-kernel design space; rather, it compares a representative set of commonly used fidelity-based constructions under a tightly controlled protocol. This framing is consistent with recent work arguing that empirical conclusions in QML are often regime-dependent and highly sensitive to protocol choices Schuld et al. (2021); Bowles et al. (2024); Schnabel and Roth (2025).

3.6 Model selection and repeated-run protocol

A major source of ambiguity in robustness studies is the selection criterion. Strong ID performance does not necessarily imply strong OOD performance, and mixing the two objectives can make results difficult to interpret. We therefore use two complementary family-internal selection modes.

Best-by-OOD.

Within each family (classical or quantum), we select the configuration with the highest mean OOD balanced accuracy. We then compare the selected classical model against the selected quantum model. This is the robustness-oriented view and constitutes the primary comparison in the paper.

Best-by-ID.

Within each family, we instead select the configuration with the highest mean ID balanced accuracy and then compare the selected classical and quantum models. This is the separability-oriented view and is reported separately to avoid conflating ID accuracy with OOD robustness.

This separation is methodological, not cosmetic. It acts as a guardrail against over-interpreting ID gains as robustness gains and helps disentangle two distinct questions: which kernel family yields the strongest in-distribution class separation, and which yields the strongest out-of-distribution generalization.

Selection is only one source of ambiguity in robustness studies; a second is result stability. Each configuration is therefore repeated across 15 runs, generated by combining five q-split seeds with three model seeds under fixed setting definitions (shift variant, master seed, and size regime). Within a setting, both classical and quantum candidates are evaluated on identical split definitions and summarized by mean $\pm$ standard deviation over the repeated runs, so that every reported comparison is variability-aware while preserving strict comparability between kernel families.

Statistical testing.

Family-level comparisons are tested formally with paired two-sided Wilcoxon signed-rank tests over the per-setting deltas (the settings are the paired units), separately for each classifier and selection view, with Holm correction applied within each classifier across the reported comparisons. We complement these tests with the per-setting effect-size proxy defined below, so that both the consistency of the sign pattern and the magnitude relative to run-to-run noise are visible.

3.7 Evaluation metrics

Our primary metric is balanced accuracy, reported separately on ID and OOD test sets:

$$\mathrm{BAcc}_{\mathrm{ID}}, \qquad \mathrm{BAcc}_{\mathrm{OOD}}.$$

Balanced accuracy is preferred because it remains informative under moderate class imbalance and is straightforward to interpret in both settings.

The ID-to-OOD degradation

$$\mathrm{drop} = \mathrm{BAcc}_{\mathrm{ID}} - \mathrm{BAcc}_{\mathrm{OOD}}$$

is recorded for every configuration and available in the artifact; because it conflates OOD behavior with the corresponding ID baseline, we treat OOD balanced accuracy as the primary inferential target and use drop only as a secondary descriptor.

To compare families after selection, we compute family deltas:

$$\Delta_{\mathrm{OOD}} = \mathrm{BAcc}^{(Q)}_{\mathrm{OOD}} - \mathrm{BAcc}^{(C)}_{\mathrm{OOD}}, \tag{3}$$

$$\Delta_{\mathrm{ID}} = \mathrm{BAcc}^{(Q)}_{\mathrm{ID}} - \mathrm{BAcc}^{(C)}_{\mathrm{ID}}. \tag{4}$$

Positive values indicate an advantage for the selected quantum model relative to the selected classical model.

Because average gains can be misleading when variability is large, we also report an effect-size-to-variability proxy:

$$\mathrm{effect} = \frac{\mu_Q - \mu_C}{\sqrt{\sigma_Q^2 + \sigma_C^2}},$$

where μ_Q, μ_C and σ_Q, σ_C are the repeated-run means and standard deviations for the metric of interest. Values greater than 1 indicate that the observed mean gain exceeds the combined variability scale, which provides a simple and interpretable robustness check on whether an apparent gain is large relative to run-to-run noise.

Probabilistic metrics.

For the Gaussian process classifier we additionally report, on both ID and OOD splits, the negative log-likelihood (log-loss), the Brier score, the expected calibration error (ECE, 15 equal-width bins), and the mean predictive entropy. These quantities characterize how the quality and honesty of predictive uncertainty change under shift: a well-behaved probabilistic model should become *more* uncertain on drifted inputs (rising entropy) while its calibration degrades as little as possible (bounded ECE increase).

3.8 Kernel-geometry descriptors

To move from measuring outcomes to explaining them, we compute geometry descriptors of the Gram matrices that every kernel induces in every setting and dimension. All descriptors are computed from the very kernel matrices consumed by the classifiers (unit diagonal by construction), so the analysis describes exactly the objects the models see.

Effective rank.

For the training Gram matrix K_{tr} with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$ normalized to $p_i = \lambda_i / \sum_j \lambda_j$, the effective rank is $\exp(-\sum_i p_i \log p_i)$ Roy and Vetterli (2007). It measures how many directions of feature space the kernel actually uses: a cosine-normalized linear kernel on low-dimensional embeddings is near rank one, while fidelity kernels can spread mass over many directions.

Centered kernel-target alignment.

For a kernel block K on a labeled split with labels $y \in \{-1,+1\}^n$, the centered kernel-target alignment (KTA) is $\langle K_c, yy^\top\rangle_F/(n\,\|K_c\|_F)$, where K_c is the doubly centered kernel Cristianini et al. (2001); Cortes et al. (2012). We evaluate KTA on the training, ID, and OOD splits separately; the ID-to-OOD change in alignment measures whether the label structure remains visible to the kernel's geometry after shift—the quantity we will call *alignment survival*.

Concentration and cross-split similarity.

We record the mean and standard deviation of off-diagonal kernel entries within the training block and of the cross blocks between each test split and the training set, together with the ratio of mean OOD-to-train and ID-to-train similarities, which measures how novel the OOD inputs look under each kernel's similarity notion.

Geometric difference.

Between each classical training kernel K_C and each quantum training kernel K_Q we compute the regularized geometric difference $g(K_C \to K_Q) = \|\sqrt{K_Q}\,(K_C + \lambda n I)^{-1}\sqrt{K_Q}\|_\infty^{1/2}$ of Huang et al. Huang et al. (2021), over a sweep of relative regularizers λ. Large values indicate that the quantum geometry can deviate substantially from what the classical kernel expresses; the quantity bounds the *potential* for a performance separation but not its sign.

Mechanistic analyses.

These descriptors feed two complementary analyses. First, a diagnostic analysis at the setting level asks which descriptors of the Best-by-OOD-selected kernels separate classical-favorable from quantum-favorable settings (Mann–Whitney AUC over settings) and correlate with the observed OOD delta. Second, a within-setting analysis asks whether geometry predicts performance across *all* kernel–dimension cells of a setting: for every setting, classifier, and q-split seed we compute the Spearman rank correlation between a descriptor (effective rank, OOD alignment) and OOD balanced accuracy over the 30–55 evaluated cells, and then examine the distribution of these correlations across settings and datasets. A descriptor that yields consistently positive within-setting correlations across datasets, drift mechanisms, and classifiers constitutes a dataset-general regularity rather than a benchmark artifact.

3.9 Experimental grid and reporting

The EMBER evaluation grid comprises 18 principal settings formed by the Cartesian product of:

- two shift variants ($m1$, $m2$),
- three master seeds,
- and three training-size regimes.

The three size regimes are:

$$(1000, 500, 500), \quad (2000, 1000, 1000), \quad (4000, 1800, 1800),$$

corresponding to $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$.

The network-flow grid applies the same structure to the three scenarios (UNSW-DoS, UNSW-Reconnaissance, ToN-IoT-Scanning), each under two shift mechanisms (m2-centroid and natural drift), three master seeds, and the same three size regimes, yielding 54 principal settings. Together the study comprises 72 principal settings, each evaluated across the dimension sweep and repeated over 15 runs, for both classifiers. The bandwidth-symmetric sweep enlarges the kernel candidate sets on the full EMBER grid and on the natural-drift network settings.

Table 1 Compact summary of the controlled experimental protocol. The repeated-run design is generated by combining five split seeds with three model seeds, yielding 15 runs per evaluated configuration under fixed setting definitions.

Component	Specification
Datasets	EMBER; UNSW-NB15 (DoS, Reconnaissance); ToN-IoT (Scanning)
Shift constructions	EMBER: $m1$, $m2$; network: m2-centroid, natural drift
Master seeds	42, 123, 999
Split seeds	42, 123, 999, 7, 2024
Model seeds	42, 123, 999
Runs per configuration	$5 \times 3 = 15$
Size regimes $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$	(1000, 500, 500), (2000, 1000, 1000), (4000, 1800, 1800)
Dimensions	$d \in \{4, 6, 8, 10, 12\}$
Classifiers	SVC (precomputed); Laplace-GP classifier (precomputed)
Shared preprocessing	MaxAbsScaler + TruncatedSVD + angle mapping
Classical-original family	Linear, RBF(scale)
Classical-extended family	+ poly(2,3), Laplacian, Matérn(3/2, 5/2), median heuristic
Quantum family	ZZ (1 rep), ZZ (2 reps), PauliXZ (1 rep), Z-map
Bandwidth sweep	RBF γ factors $\{0.1, 0.3, 1, 3, 10\}$; angle scales $\{0.5, 1, 2\}$
Selection views	Best-by-OOD, Best-by-ID (family-internal, on run means)
Statistical testing	Paired Wilcoxon over settings + Holm; effect proxy
Principal settings	EMBER 2×3×3 = 18; network 3×2×3×3 = 54; total 72

For clarity and reproducibility, Table 1 summarizes the main experimental protocol used throughout the paper, including the shift constructions, controlled seed structure, size regimes, dimensionality sweep, kernel families, classifiers, and repeated-run design.

Within each setting, we evaluate both kernel families across the shared preprocessing grid and the dimension sweep $d \in \{4, 6, 8, 10, 12\}$. This produces both a setting-level comparison, used for the main Best-by-OOD and Best-by-ID analyses, and a broader dimension-level view used to characterize how representation size interacts with kernel-family behavior.

For reproducibility, split definitions are fixed within each setting, random seeds are controlled, and all key quantities are reported as mean ± standard deviation across repeated runs. Throughout the paper, we deliberately avoid claims of universal dominance or complexity-theoretic quantum advantage. The object of study is narrower and more empirical: under a fixed classifier and a fixed evaluation protocol, does changing only the kernel alter robustness under distribution shift in security-relevant tasks—and which property of the kernel is responsible?

4 Results

4.1 Overview

We report results under the controlled kernel-swap protocol described in Section 3. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a setting; the only varying component is the kernel, and every number below is a mean over 15 repeated runs. The results are organized as an argument in four movements. First, on EMBER, the quantum family is compared against the customary linear+RBF baselines (Section 4.2). Second, the classical reference set is strengthened to the extended family, which changes the verdict on one modality (Section 4.3). Third, both families receive the same bandwidth-tuning freedom, which changes it back (Section 4.4). Fourth, the protocol is repeated on three network-flow scenarios including natural drift (Section 4.5). Section 4.6 then explains the entire pattern with a single geometric mechanism, and Section 4.7 examines predictive uncertainty under shift.

4.2 EMBER versus the customary baselines: a near-uniform quantum advantage

We begin with the comparison that dominates the literature: fidelity-based quantum kernels against linear and RBF baselines. For each setting and classifier, family-internal Best-by-OOD selection picks the configuration with the highest mean OOD balanced accuracy on the run means, and the selected representatives are compared directly. Table 2 (upper panel, first and third rows) reports the outcome.

The advantage is close to uniform. Under SVC the selected quantum kernel improves OOD balanced accuracy in 17 of 18 settings (mean $\Delta_{\text{OOD}} = +0.037$, Holm-adjusted Wilcoxon $p = 7.6 \times 10^{-5}$), with the gain exceeding the combined run-to-run variability scale in 12 settings; under the GP classifier the corresponding figures are 15 of 18, +0.028, $p = 2.1 \times 10^{-4}$, and 11 settings. The few classical-favorable settings are marginal: all occur at the largest training size under the $m1$ construction, with $|\Delta_{\text{OOD}}| \leq 0.010$ and effect sizes below one, i.e., within run-to-run noise. The dimension-level view confirms that this is not a best-of-grid artifact: comparing the best quantum against the best classical candidate separately at every setting–dimension cell, the quantum family is ahead in 88 of 90 cells under SVC and 85 of 90 under the GP classifier (mean $\Delta_{\text{OOD}} \approx +0.05$ in both cases).

Under Best-by-ID selection (lower panel), the quantum family improves ID balanced accuracy in all 18 settings under both classifiers, with mean gains of +0.078 (SVC) and +0.065 (GPC) and every setting exceeding the variability scale. We keep the two views separate throughout: strong ID separability is not evidence of robustness, and the following subsections show that the two properties respond very differently to stronger classical baselines. Figure 1 previews the resulting arc: the per-setting OOD deltas of this subsection (first column of each panel) collapse toward zero against the extended family (second column), partially recover under bandwidth symmetry (third), and re-emerge on network flows (fourth).

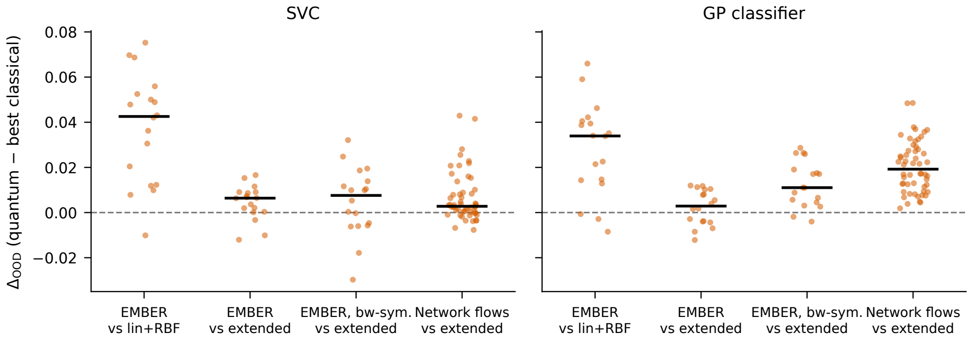

Fig. 1 Per-setting Best-by-OOD deltas (quantum minus best classical, mean over 15 runs) across the four movements of the study, for SVC (left) and the GP classifier (right). Black bars mark medians. Against linear+RBF the quantum advantage is large and nearly uniform; the extended classical family collapses it on EMBER; symmetric bandwidth tuning partially restores it; on network flows (54 settings, including natural drift) it persists against the extended family, uniformly so under the GP classifier.

Table 2 EMBER headline results (18 settings, 15 runs each). Within each classifier (GPC = Laplace Gaussian process classifier), family-internal selection compares the best quantum candidate against the best candidate of the stated classical reference set. p-values are two-sided paired Wilcoxon over settings, Holm-corrected within classifier; the last column counts settings whose quantum-favorable gain exceeds the combined run-to-run variability scale.

Selection	Classifier	Classical ref.	Q wins	Mean Δ	Median Δ	p (Holm)	Eff.>1
Best-by-OOD	SVC	linear+RBF	17/18	+0.0374	+0.0426	7.6×10^{-5}	12
	SVC	extended	15/18	+0.0044	+0.0064	0.03	1
	GPC	linear+RBF	15/18	+0.0283	+0.0339	2.1×10^{-4}	11
	GPC	extended	11/18	+0.0023	+0.0029	0.26	0
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794	3.1×10^{-5}	18
	SVC	extended	18/18	+0.0149	+0.0152	3.1×10^{-5}	11
	GPC	linear+RBF	18/18	+0.0649	+0.0664	3.1×10^{-5}	18
	GPC	extended	18/18	+0.0192	+0.0166	3.1×10^{-5}	11

4.3 The extended classical family neutralizes the OOD advantage on EMBER

The comparison changes character when the classical family is strengthened. Against the extended set—which adds polynomial, Laplacian, and Matérn kernels with train-only median-heuristic scales—the quantum OOD advantage on EMBER collapses by an order of magnitude (Table 2, second and fourth rows): the mean delta falls from

+0.037 to +0.004 under SVC and from +0.028 to +0.002 under the GP classifier, the gain exceeds the variability scale in at most one setting, and under the GP classifier the comparison is no longer statistically significant (Holm-adjusted $p = 0.26$). The winner of this neutralization is overwhelmingly one kernel: the median-heuristic Laplacian is the Best-by-OOD selection of the extended classical family in 13 of 18 settings under SVC and 14 of 18 under the GP classifier, almost always at the largest embedding dimension.

Two aspects of this result deserve emphasis. First, it is an honest negative result for the quantum-advantage narrative as usually framed: the OOD gains reported against linear and RBF baselines—here and, we suspect, elsewhere in the literature—measure the weakness of those baselines as much as the strength of the quantum family. Second, the *ID* advantage does not collapse: even against the extended family, the quantum kernels improve ID balanced accuracy in 18 of 18 settings under both classifiers ($p = 3.1 \times 10^{-5}$), with the gain exceeding the variability scale in 11 settings. Whatever the quantum feature maps contribute to in-distribution separability, no classical kernel in our set replicates it; what the Laplacian replicates is specifically the out-of-distribution behavior. Section 4.6 shows that this is not a coincidence: the Laplacian is precisely the classical kernel whose Gram-matrix geometry most resembles that of the fidelity maps.

4.4 Bandwidth symmetry restores the advantage

The comparison of Section 4.3 pins every quantum feature map to the canonical $[0, \pi]$ encoding while granting the classical family kernels whose length scales adapt to the data. Bandwidth, however, is as decisive for fidelity kernels as it is for the RBF kernel Shaydulin and Wild (2022); Canatar et al. (2023). When both families receive the same tuning axis—RBF gamma factors $\{0.1, 0.3, 1, 3, 10\}$ against quantum angle scales $\{0.5, 1, 2\}$—the picture inverts again. Under the GP classifier, the quantum family beats the bandwidth-swept extended classical family in 16 of 18 EMBER settings (mean $\Delta_{\text{OOD}} = +0.013$, Holm-adjusted $p = 7.6 \times 10^{-5}$); under SVC the comparison is positive but not significant (11–12 of 18, $p \approx 0.15$). The mechanism is visible in the selections: the angle-scaled variants dominate the quantum family's Best-by-OOD choices (scales 0.5 and 2 are selected far more often than the canonical encoding), confirming that the fixed-bandwidth comparison had the quantum family playing with one hand tied. On the network-flow scenarios under natural drift, the same symmetric sweep leaves the quantum advantage intact and significant under both classifiers (24 of 27 settings, Holm-adjusted $p = 5.2 \times 10^{-5}$ for the GP classifier; 19 of 27, $p = 9.8 \times 10^{-4}$ for SVC).

The methodological moral generalizes beyond this study: fixed-encoding quantum kernels versus tuned classical kernels is not a protocol-fair comparison, and the bias runs *against* the quantum family. Once tuning freedom is symmetric, the extended classical family no longer neutralizes the quantum OOD advantage.

4.5 Network flows: the advantage survives the extended family and peaks under natural drift

The 54 network-flow settings test whether the EMBER conclusions are properties of one dataset family or of the kernels. They are neither, exactly: the network results are *stronger* for the quantum family than EMBER's (Table 3). Against linear+RBF, the selected quantum kernel improves OOD balanced accuracy in 54 of 54 settings under both classifiers (Holm-adjusted $p = 6.5 \times 10^{-10}$), with the gain exceeding the variability scale in 35 (SVC) and 44 (GPC) settings. More importantly, against the *extended* classical family—the set that neutralized the advantage on EMBER—the quantum family remains ahead in 54 of 54 settings under the GP classifier (mean $\Delta_{\text{OOD}} = +0.020$, $p = 6.5 \times 10^{-10}$, effect$>$ 1 in 33) and in 41 of 54 under SVC

Table 3 Network-flow headline results (54 settings across UNSW-NB15 DoS/Reconnaissance and ToN-IoT Scanning, two shift mechanisms, 15 runs each). Columns as in Table 2.

Selection	Classifier	Classical ref.	Q wins	Mean Δ	Median Δ	p (Holm)	Eff.> 1
Best-by-OOD	SVC	linear+RBF	54/54	+0.0256	+0.0264	6.5×10^{-10}	35
	SVC	extended	41/54	+0.0066	+0.0028	3.2×10^{-5}	7
	GPC	linear+RBF	54/54	+0.0344	+0.0338	6.5×10^{-10}	44
	GPC	extended	54/54	+0.0204	+0.0192	6.5×10^{-10}	33
Best-by-ID	SVC	linear+RBF	54/54	+0.0210	+0.0104	6.5×10^{-10}	29
	SVC	extended	39/54	+0.0036	+0.0028	1.5×10^{-4}	11
	GPC	linear+RBF	53/54	+0.0188	+0.0100	6.5×10^{-10}	24
	GPC	extended	40/54	+0.0057	+0.0044	7.9×10^{-7}	14

($p = 3.2 \times 10^{-5}$). The Laplacian is again the dominant classical challenger, and again it is not enough.

The breakdown by shift mechanism sharpens the picture. Under SVC, the quantum-versus-extended comparison is essentially tied on the synthetic centroid construction (15 of 27 settings) but nearly uniform under natural drift (26 of 27): the advantage is largest precisely on the drift mechanism that was not constructed by us and most resembles deployment conditions. Under the GP classifier the advantage is uniform on both mechanisms (27 of 27 each). Read jointly with Section 4.3, the regime map is: on EMBER's synthetic covariate extremes a heavy-tailed classical kernel can match the quantum family at fixed bandwidth; on drifting network traffic, under a probabilistic classifier, nothing in our classical set does.

4.6 One geometric mechanism explains the pattern

The results so far pose a puzzle with three faces: why do quantum kernels beat linear and RBF baselines so uniformly, why does the Laplacian—alone among our classical kernels—catch them on EMBER, and why does natural network drift favor them again? The geometry descriptors of Section 3.8 answer all three at once.

Table 4 Within-setting Spearman correlation between kernel-geometry descriptors and OOD balanced accuracy, summarized per dataset and classifier over all setting-seed units (five q-split seeds per setting; 30–55 kernel–dimension cells per unit). "> 0" is the fraction of units with a positive correlation.

Dataset	Classifier	Units	Effective rank median ρ	Effective rank > 0	KTA on OOD median ρ	KTA on OOD > 0
EMBER	GP classifier	90	0.78	99%	0.68	89%
EMBER	SVC	90	0.83	98%	0.65	89%
ToN-IoT Scanning	GP classifier	90	0.84	100%	0.54	99%
ToN-IoT Scanning	SVC	90	0.75	100%	0.59	99%
UNSW-NB15 DoS	GP classifier	90	0.55	98%	0.51	96%
UNSW-NB15 DoS	SVC	90	0.55	90%	0.42	96%
UNSW-NB15 Recon.	GP classifier	90	0.26	98%	0.57	94%
UNSW-NB15 Recon.	SVC	90	0.19	83%	0.46	99%

A cross-dataset regularity.

Within every setting we correlate, across all evaluated kernel–dimension cells, each geometry descriptor with OOD balanced accuracy (Table 4 and Figure 2). The alignment-survival quantity—centered kernel-target alignment evaluated on the OOD split—predicts OOD accuracy within settings essentially everywhere: the within-setting Spearman correlation is positive in 89–99% of the 360 setting-seed units, with median ρ between 0.42 and 0.74, across all four datasets, both classifiers, and all three drift mechanisms. Effective rank of the training kernel—a train-only quantity—is an equally strong predictor on EMBER and ToN-IoT (median $\rho \approx$ 0.75–0.84), moderate on UNSW-DoS, and weak (though still predominantly positive) on UNSW-Reconnaissance. The natural reading is that high effective rank is the structural precondition—a kernel whose geometry uses only one or two directions has nothing with which to track label structure into the tails—while alignment survival is the proximate predictor of OOD performance.

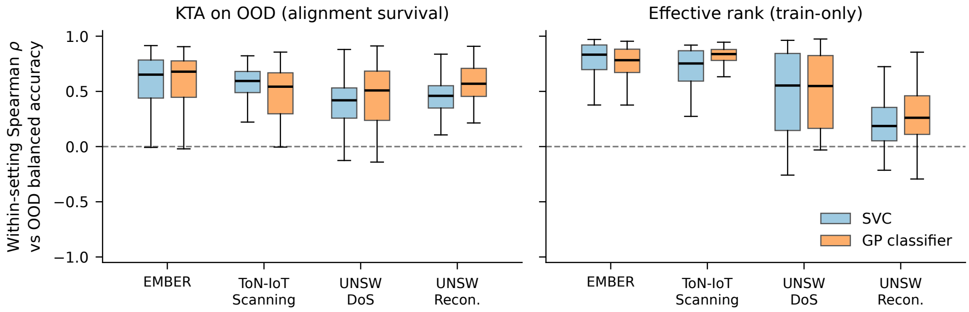

Fig. 2 The mechanism is a cross-dataset regularity. Distribution of within-setting Spearman correlations between each geometry descriptor and OOD balanced accuracy, over all setting-seed units, per dataset and classifier. Alignment survival (left) is positively correlated with OOD accuracy essentially everywhere; effective rank (right) is strong on EMBER and ToN-IoT and weaker, though still predominantly positive, on the UNSW scenarios.

A dose-response gradient within the classical family.

The mechanism makes a prediction that can be tested without any quantum kernel at all: classical kernels with higher effective rank should retain label alignment better under shift. Table 5 and Figure 3 confirm a clean monotone gradient across the seven classical kernels (18 EMBER settings $\times$ 5 dimensions each): as mean effective rank rises from 1.19 (linear) through 2.76 (RBF) and 6–8 (Matérn) to 17.7 (Laplacian), the mean ID-to-OOD change in alignment rises monotonically from -0.017 to $+0.019$, and the fraction of cells where alignment *improves* under shift doubles from 32% to 66%. This explains Section 4.3 precisely: the Laplacian is the classical kernel that shares the quantum maps' geometric signature, and it is exactly the kernel that neutralizes their OOD advantage. Conversely, the fidelity maps' near-uniform advantage over linear and RBF baselines is an advantage over the two lowest-rank geometries in the study.

What the geometry does not do.

Two negative results complete the mechanistic picture, and both are informative. First, the geometric difference of Huang et al. Huang et al. (2021)—the quantity a reader versed in the quantum-kernel literature would reach for first—does not predict which family wins a setting; at moderate regularization it even correlates *negatively* with the

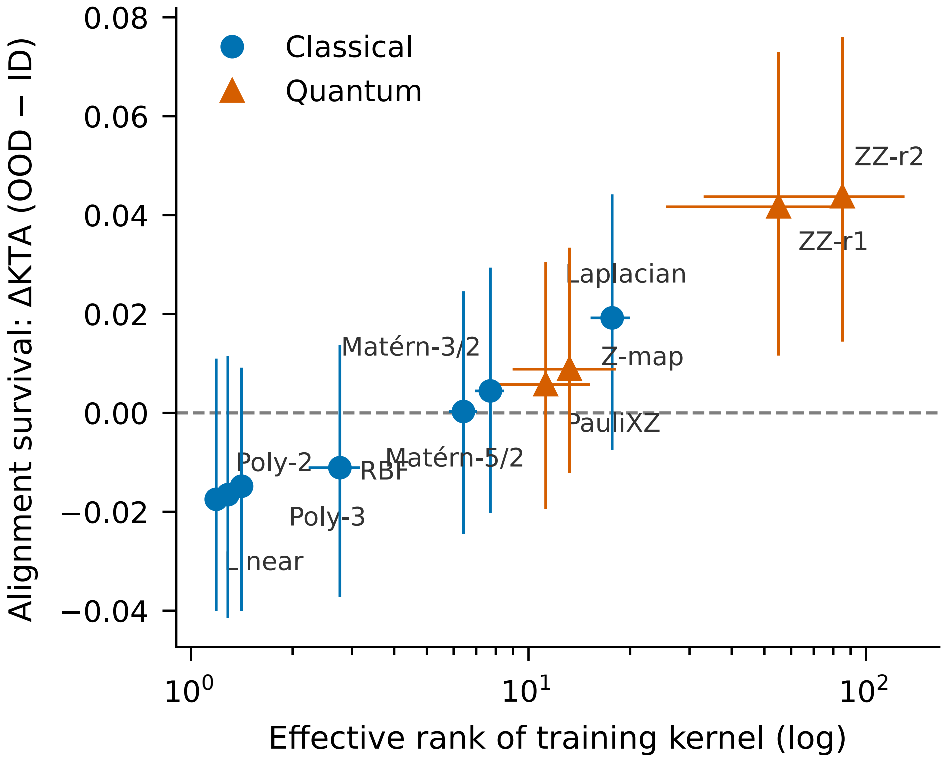

Fig. 3 Effective rank versus alignment survival for every kernel in the study (means over the 18 EMBER settings and 5 dimensions; bars span interquartile ranges). The relationship is monotone across both families: the classical kernels that retain alignment under shift are exactly the heavy-tailed ones, and the fidelity maps occupy the high-rank end of the same continuum.

observed OOD delta. This is consistent with its definition: g bounds the potential for a separation between the families, not its sign, so a quantum kernel can differ greatly from the classical reference and be worse. Second, we found no reliable fully a-priori selection rule: choosing the quantum configuration by training-set alignment alone does not identify the settings where the family will win. The geometry explains and diagnoses; it does not yet provide a deployment-time oracle, and we flag this honestly as an open problem.

Table 5 Dose-response gradient in the classical family (18 EMBER settings × 5 dimensions, q-split seed 42). Higher effective rank of the training Gram matrix is associated with better survival—indeed improvement—of kernel-target alignment from ID to OOD.

Kernel	Mean eff. rank	Mean ΔKTA (OOD−ID)	KTA improves
Linear	1.19	−0.017	32%
Polynomial (2)	1.29	−0.017	32%
Polynomial (3)	1.41	−0.015	38%
RBF (scale)	2.76	−0.011	38%
Matérn 5/2	6.41	+0.000	46%
Matérn 3/2	7.71	+0.004	47%
Laplacian	17.69	+0.019	66%

4.7 Predictive uncertainty under shift

The GP classifier's calibrated probabilities add a complementary robustness lens. Averaged over Best-by-OOD selections, predictive entropy rises from ID to OOD for every family and dataset group—the qualitative behavior one wants from a probabilistic detector facing drift—and rises most for the quantum kernels (e.g., +0.029 versus +0.014–+0.018 for the classical families on EMBER; +0.072 versus +0.041–+0.061 on network flows). Calibration degrades moderately for all families (ECE increases of +0.02–+0.05), with the quantum family's ECE increase slightly larger than the classical families'. In other words, the quantum kernels' OOD accuracy advantage does not come with hidden overconfidence—their uncertainty grows under shift at least as fast as the classical kernels'—but neither do they calibrate better; the advantage is in the decision geometry, not in the probability calibration.

5 Discussion

5.1 What is, and is not, a quantum advantage here

The study's central interpretive move is to replace the question "do quantum kernels beat classical kernels under shift?" with "which property of a kernel governs its behavior under shift, and who has it?". Read through that lens, the results decompose cleanly. The near-uniform advantage over linear and RBF baselines is real, replicable across four datasets and two classifiers, and statistically unambiguous—but it is an advantage of *high-effective-rank, alignment-preserving geometry* over the two lowest-rank geometries in the study, a property the fidelity maps possess natively rather than exclusively. The Laplacian result makes this precise: one classical kernel with the same geometric signature closes the OOD gap on EMBER, so the property, not the physics, is doing the work there. What remains after every control we could devise—extended baselines, symmetric bandwidth tuning, a probabilistic classifier, natural drift—is a genuine residual: the ID-separability advantage survives everything on EMBER, and the OOD advantage survives everything on drifting network traffic. Those two residuals are, on present evidence, the defensible quantum contribution.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are only as credible as the protocol used to obtain them Quiñonero-Candela et al. (2008); Gulrajani and Lopez-Paz (2021); Koh et al. (2021); Yao et al. (2022). Our results add a QML-specific corollary: the composition of the classical reference set and the symmetry of tuning freedom are not implementation details but the difference between a significant advantage, a null result, and a significant advantage again (Sections 4.2–4.4). Any single one of those three verdicts, reported alone, would have been misleading.

5.2 The ID–OOD dissociation is informative

The contrast between the two selection views is sharper in this study than in its predecessors, and it now has an interpretation. The quantum ID-separability advantage is universal (18 of 18 EMBER settings against every classical reference, with effects exceeding run-to-run noise in most) and is not replicated by any classical kernel we evaluate, including the Laplacian. The OOD advantage, by contrast, is contingent: it depends on the reference family, the bandwidth protocol, and the drift mechanism. This dissociation is exactly what the mechanism predicts. In-distribution separability rewards expressive geometry per se; robustness rewards geometry whose label alignment *survives* the shift, which is a joint property of the kernel and the shift mechanism. A kernel family can therefore dominate the first criterion while merely competing on the second—and mistaking one for the other is precisely the error the two-view protocol is designed to prevent Gulrajani and Lopez-Paz (2021).

5.3 The mechanism in context

The alignment-survival regularity connects several strands of literature that have so far run in parallel. The bandwidth results extend the findings of Shaydulin and Wild Shaydulin and Wild (2022) and Canatar et al. Canatar et al. (2023) from in-distribution generalization to robustness under shift, and give them a comparative twist: bandwidth is not merely important for quantum kernels, it is important *symmetrically*, and asymmetric tuning protocols systematically bias quantum-versus-classical comparisons. The failure of the geometric difference Huang et al. (2021) to predict OOD outcomes sharpens its interpretation rather than contradicting it: g was derived as a bound on the potential for in-distribution separations, and our results show empirically that potential-for-difference and benefit-under-shift are distinct quantities—large g is compatible with large losses. Finally, the dose-response gradient within the classical family (Table 5) grounds the mechanism independently of anything quantum: kernels with heavier spectral tails hold their label alignment into the tails of the data distribution, a statement about kernels generally, for which the quantum maps are simply the most extreme instances in our study.

We emphasize what the mechanism does not yet deliver. It is explanatory and diagnostic—given the Gram matrices of a setting, it accounts for which kernels will generalize under the shift—but our attempts to convert it into an a-priori selection rule from training data alone failed (Section 4.6). Alignment on OOD inputs requires OOD labels; effective rank is train-only but is a precondition, not a guarantee. Turning alignment survival into a deployable model-selection signal, perhaps via unlabeled OOD data or distributionally robust surrogates, is in our view the most interesting open problem this study raises.

5.4 Implications for security machine learning practice

From a practical perspective, the paper suggests three takeaways for security ML. The first is substantive: for OOD-sensitive kernel selection, prefer kernels whose label-aligned geometry survives shift. Concretely, if quantum kernels are unavailable or too costly, a median-heuristic Laplacian kernel is a far stronger robustness baseline than the RBF kernel that pipelines default to—on our EMBER settings it recovers most of what the fidelity maps offer. Where quantum kernels are on the table, they warrant serious consideration for drifting-traffic workloads, where their advantage was largest and most uniform in our study.

The second is methodological. Independently of the preferred kernel family, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed, and swap only the kernel;
2. define explicit OOD splits tied to interpretable shift mechanisms, including at least one natural drift mechanism where the data permit;
3. give every family the same hyperparameter freedom, bandwidth included;
4. repeat runs, report variability, and test family-level claims formally;
5. and separate model-selection criteria according to the actual objective (robustness vs. separability).

Security ML frequently suffers from comparisons in which model class, tuning effort, and evaluation conditions change simultaneously; the three-verdict sequence of

Sections 4.2–4.4 is a concrete demonstration of how much the conclusion depends on getting this right.

The third concerns diagnostics: the geometry descriptors are cheap to compute from Gram matrices that the pipeline already builds, and the alignment-survival analysis can be run post hoc on any deployed kernel system with a labeled drift sample. Practitioners should nonetheless resist reading these results as a blanket recommendation to replace classical kernels; the responsible recommendation is to test the candidate families under the expected shift regime, using exactly the kind of protocol described here.

5.5 Limitations and threats to validity

Several limitations define the scope of our conclusions.

Shift operationalization.

Our OOD splits cover three reproducible mechanisms—within-class sparsity extremes, train-geometry extremes, and natural regime drift on network traffic—but not the full space of real-world drift processes. Explicit temporal splits on timestamped malware, family-based drift, and adversarially induced shift remain outside the study's scope, and the alignment-survival regularity should be re-tested under them rather than assumed.

Kernel and encoding coverage.

The classical family now spans seven kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom, but neither candidate set is exhaustive: the quantum side is restricted to four commonly used fidelity feature maps at low qubit counts, and trained or task-adapted encodings, projected kernels, and deeper maps are not evaluated. The mechanism gives a concrete prediction for such extensions—what should matter is their Gram-matrix geometry, not their provenance—which future work can falsify.

Selection over candidate sets.

Best-by-OOD and Best-by-ID are selection procedures over finite candidate sets, applied identically to both families and evaluated on run means with formal paired testing. Residual sensitivity to the candidate space and search granularity nonetheless remains, and nested or cross-validated selection protocols would strengthen the analysis further.

Simulation, not hardware.

All quantum kernels are computed by exact statevector simulation. The study therefore characterizes the geometry of fidelity kernels, not the behavior of noisy devices: finite sampling of kernel entries, hardware noise, and error mitigation would all perturb the Gram matrices, and their effect on alignment survival is unknown. We make no claim about hardware feasibility, resource advantage, or quantum speedup; at the dimensions studied, the fidelity kernels are also classically simulable, which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats.

Four benchmark scenarios across two modalities is a substantial widening relative to single-dataset studies, but security is broader still: dynamic malware analysis, encrypted traffic, and non-security modalities remain untested. Within the network scenarios, the feature sets inherited from the public benchmarks retain port numbers and protocol indicators, which are legitimate but known shortcut-prone features in NIDS; because every kernel receives identical features, this affects absolute accuracies, not the family comparisons. Finally, the mechanism's effective-rank component is itself regime-dependent (strong on EMBER and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than smooth over.

5.6 Conservative interpretation

In light of these limitations and the broader QML benchmarking literature, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or hardware relevance. We claim, with formal statistical support, three narrower things: that fidelity-based quantum kernels beat the customary baselines under shift essentially wherever we looked; that this fact is explained by a geometric property a well-chosen classical kernel can partially share; and that under symmetric protocols the quantum family retains a significant OOD advantage on drifting network traffic and a universal ID-separability advantage on EMBER. This regime-aware view is consistent with recent calls for careful benchmarking in QML Bowles et al. (2024); Schnabel and Roth (2025) and, we believe, is scientifically stronger than either an unqualified advantage claim or an unqualified debunking: it says what the advantage is made of, where it lives, and how to check for it.

6 Conclusion

This paper asked why kernel families behave differently under distribution shift, using the quantum-versus-classical comparison as both motivation and stress test. A controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and tuning freedom held fixed while only the kernel varies—was instantiated on four security benchmarks spanning two modalities and three drift mechanisms, two classifier families including a probabilistic one, and 72 principal settings with 15 repeated runs and formal paired testing.

The empirical arc has four movements. Fidelity-based quantum kernels beat the customary linear and RBF baselines under shift nearly everywhere. A stronger classical family neutralizes that advantage on EMBER, revealing that the operative property is geometric rather than quantum. Symmetric bandwidth tuning restores it, revealing that the neutralization itself depended on an asymmetric protocol. And on drifting network traffic the advantage is uniform under the probabilistic classifier, while the in-distribution separability advantage survives every control on EMBER. A single mechanism accounts for the whole pattern: kernels whose training geometry has high effective rank retain their kernel-target alignment under shift, and alignment survival predicts out-of-distribution accuracy within settings in 89–99% of 360 setting-seed units across all four datasets. The mechanism also explains the exceptions—the competitive classical kernels are exactly those sharing the geometric signature—and makes falsifiable predictions for kernel constructions we did not test.

The contribution is therefore threefold. Empirically, the study provides the broadest controlled, variability-aware regime map of quantum-versus-classical kernel behavior under shift that we are aware of, including the honest negative results (the Laplacian neutralization, the non-predictiveness of the geometric difference, the absence of an a-priori selection rule). Mechanistically, it converts an observation into an explanation with cross-dataset support. Methodologically, it leaves behind a fully reproducible protocol—and a cautionary demonstration that the quantum-versus-classical verdict flips twice as the protocol is made progressively fairer, which we offer as a concrete standard for future QML benchmarking.

The most important open problems are two. Practically, turning alignment survival into a deployment-time selection signal without OOD labels would convert the diagnosis into a tool. Theoretically, the dose-response relationship between spectral tails and alignment survival calls for a formal account: under which shift models does high effective rank provably protect kernel-target alignment? Answering either question would benefit classical and quantum kernel methods alike—which is, in the end, the point: the productive question is not whether one family dominates, but which similarity geometry aligns best with the shift regime one actually faces.

Appendix A Additional Results

This appendix reports the full setting-level Best-by-OOD comparisons between the selected quantum kernel and the best kernel of the *extended* classical family (the strongest comparison in the study), for both classifiers and all 72 principal settings. Every value is a mean ± standard deviation over 15 repeated runs; Effect denotes $\Delta_{\mathrm{OOD}}/\sqrt{\sigma_C^2+\sigma_Q^2}$. Size labels denote $(n_{\mathrm{train}}, n_{\mathrm{ID}}, n_{\mathrm{OOD}})$: S = (1k, 500, 500), M = (2k, 1k, 1k), L = (4k, 1.8k, 1.8k). Shift labels for the network scenarios abbreviate scenario/mechanism: UD = UNSW-NB15 DoS, UR = UNSW-NB15 Reconnaissance, TS = ToN-IoT Scanning; /m2 is the centroid construction and /nat is natural drift. Quantum kernel names carry the angle scale after `@` where it differs from 1. All tables are generated directly from the public artifact by `scripts/reporting/make_v2_tables.py`.

Table A1 EMBER, SVC: full Best-by-OOD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
m1	42	S	Laplacian (d4)	PauliXZ (d12)	0.756 ± 0.007	0.758 ± 0.020	$+0.002$	$+0.09$
m1	42	M	Laplacian (d12)	ZZ-r1 (d12)	0.753 ± 0.019	0.761 ± 0.007	$+0.009$	$+0.42$
m1	42	L	Laplacian (d12)	ZZ-r1 (d12)	0.750 ± 0.014	0.757 ± 0.008	$+0.006$	$+0.38$
m1	123	S	Laplacian (d12)	PauliXZ (d12)	0.762 ± 0.019	0.750 ± 0.020	-0.012	-0.43
m1	123	M	Laplacian (d12)	ZZ-r1 (d10)	0.760 ± 0.027	0.756 ± 0.029	-0.003	-0.08
m1	123	L	Laplacian (d12)	ZZ-r1 (d10)	0.751 ± 0.019	0.753 ± 0.014	$+0.002$	$+0.09$
m1	999	S	Laplacian (d12)	PauliXZ (d12)	0.767 ± 0.028	0.768 ± 0.022	$+0.000$	$+0.00$
m1	999	M	Matérn-5/2 (d12)	ZZ-r1 (d12)	0.755 ± 0.015	0.762 ± 0.013	$+0.007$	$+0.34$
m1	999	L	RBF (d12)	ZZ-r1 (d10)	0.763 ± 0.024	0.753 ± 0.005	-0.010	-0.42
m2	42	S	Laplacian (d10)	ZZ-r1 (d10)	0.762 ± 0.012	0.763 ± 0.018	$+0.000$	$+0.02$
m2	42	M	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.012	0.784 ± 0.009	$+0.017$	$+1.10$
m2	42	L	Laplacian (d12)	ZZ-r2 (d12)	0.783 ± 0.008	0.795 ± 0.011	$+0.012$	$+0.85$
m2	123	S	Laplacian (d12)	ZZ-r1 (d12)	0.762 ± 0.023	0.771 ± 0.019	$+0.009$	$+0.30$
m2	123	M	Laplacian (d12)	ZZ-r1 (d12)	0.775 ± 0.017	0.783 ± 0.013	$+0.007$	$+0.35$
m2	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.785 ± 0.011	0.794 ± 0.007	$+0.009$	$+0.71$
m2	999	S	Laplacian (d12)	ZZ-r1 (d12)	0.769 ± 0.014	0.775 ± 0.022	$+0.006$	$+0.25$
m2	999	M	Laplacian (d10)	ZZ-r2 (d12)	0.776 ± 0.016	0.780 ± 0.020	$+0.004$	$+0.14$
m2	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.787 ± 0.013	0.803 ± 0.009	$+0.015$	$+0.97$

Table A2 EMBER, GP classifier: full Best-by-OOD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
m1	42	S	Laplacian (d12)	PauliXZ (d12)	0.767 ± 0.018	0.760 ± 0.017	-0.007	-0.29
m1	42	M	Laplacian (d12)	ZZ-r1 (d12)	0.765 ± 0.021	0.766 ± 0.011	$+0.002$	$+0.07$
m1	42	L	RBF (d10)	PauliXZ (d12)	0.764 ± 0.015	0.761 ± 0.011	-0.003	-0.16
m1	123	S	Laplacian (d12)	PauliXZ (d12)	0.765 ± 0.016	0.753 ± 0.025	-0.012	-0.41
m1	123	M	Laplacian (d12)	PauliXZ (d12)	0.775 ± 0.035	0.772 ± 0.023	-0.004	-0.09
m1	123	L	Matérn-5/2 (d10)	ZZ-r2 (d12)	0.758 ± 0.017	0.754 ± 0.014	-0.004	-0.18
m1	999	S	Laplacian (d12)	PauliXZ (d12)	0.766 ± 0.028	0.762 ± 0.027	-0.004	-0.11
m1	999	M	Matérn-5/2 (d12)	Z-map (d12)	0.768 ± 0.016	0.776 ± 0.009	$+0.008$	$+0.43$
m1	999	L	RBF (d12)	ZZ-r1 (d8)	0.763 ± 0.009	0.754 ± 0.008	-0.008	-0.74
m2	42	S	Laplacian (d12)	ZZ-r1 (d12)	0.762 ± 0.024	0.764 ± 0.015	$+0.002$	$+0.06$
m2	42	M	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.013	0.778 ± 0.008	$+0.010$	$+0.68$
m2	42	L	Laplacian (d12)	ZZ-r2 (d12)	0.783 ± 0.008	0.793 ± 0.009	$+0.010$	$+0.85$
m2	123	S	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.025	0.772 ± 0.015	$+0.005$	$+0.19$
m2	123	M	Laplacian (d12)	ZZ-r1 (d12)	0.772 ± 0.020	0.783 ± 0.015	$+0.011$	$+0.44$
m2	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.780 ± 0.010	0.791 ± 0.010	$+0.012$	$+0.83$
m2	999	S	Laplacian (d12)	ZZ-r1 (d10)	0.767 ± 0.021	0.775 ± 0.021	$+0.008$	$+0.27$
m2	999	M	Laplacian (d12)	ZZ-r2 (d12)	0.776 ± 0.016	0.780 ± 0.025	$+0.004$	$+0.13$
m2	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.786 ± 0.013	0.798 ± 0.011	$+0.012$	$+0.70$

Table A3 Network flows, SVC: full Best-by-OOD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
TS/m2	42	S	Laplacian (d12)	ZZ-r1 (d6)	0.941 ± 0.013	0.954 ± 0.009	$+0.014$	$+0.90$
TS/m2	42	M	Laplacian (d12)	ZZ-r1 (d10)	0.941 ± 0.014	0.958 ± 0.005	$+0.017$	$+1.15$
TS/m2	42	L	Laplacian (d12)	ZZ-r1 (d8)	0.959 ± 0.009	0.964 ± 0.006	$+0.005$	$+0.44$
TS/m2	123	S	Laplacian (d12)	ZZ-r1 (d8)	0.943 ± 0.017	0.959 ± 0.011	$+0.015$	$+0.75$
TS/m2	123	M	Laplacian (d12)	ZZ-r1 (d6)	0.954 ± 0.007	0.958 ± 0.006	$+0.003$	$+0.35$
TS/m2	123	L	Laplacian (d10)	ZZ-r1 (d8)	0.955 ± 0.003	0.959 ± 0.003	$+0.003$	$+0.72$
TS/m2	999	S	Laplacian (d10)	ZZ-r1 (d10)	0.936 ± 0.011	0.957 ± 0.003	$+0.021$	$+1.80$
TS/m2	999	M	Laplacian (d12)	ZZ-r1 (d8)	0.954 ± 0.009	0.962 ± 0.007	$+0.008$	$+0.72$
TS/m2	999	L	Laplacian (d12)	ZZ-r1 (d10)	0.965 ± 0.004	0.969 ± 0.003	$+0.003$	$+0.71$
TS/nat	42	S	Laplacian (d8)	ZZ-r1 (d10)	0.987 ± 0.003	0.988 ± 0.005	$+0.001$	$+0.10$
TS/nat	42	M	Laplacian (d10)	ZZ-r1 (d12)	0.989 ± 0.004	0.990 ± 0.002	$+0.001$	$+0.24$
TS/nat	42	L	Laplacian (d12)	Z-map (d12)	0.989 ± 0.005	0.991 ± 0.003	$+0.002$	$+0.31$
TS/nat	123	S	Laplacian (d8)	ZZ-r1 (d12)	0.990 ± 0.005	0.992 ± 0.005	$+0.002$	$+0.36$
TS/nat	123	M	Laplacian (d12)	Z-map (d10)	0.989 ± 0.003	0.992 ± 0.002	$+0.003$	$+0.61$
TS/nat	123	L	Laplacian (d10)	Z-map (d12)	0.990 ± 0.002	0.991 ± 0.003	$+0.002$	$+0.40$
TS/nat	999	S	Laplacian (d12)	Z-map (d12)	0.979 ± 0.006	0.985 ± 0.002	$+0.006$	$+1.05$
TS/nat	999	M	Laplacian (d10)	Z-map (d10)	0.990 ± 0.002	0.991 ± 0.004	$+0.002$	$+0.38$
TS/nat	999	L	Laplacian (d10)	Z-map (d12)	0.992 ± 0.002	0.992 ± 0.003	$+0.000$	$+0.12$
UD/m2	42	S	Laplacian (d12)	PauliXZ (d12)	0.920 ± 0.016	0.919 ± 0.014	-0.001	-0.06
UD/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.934 ± 0.017	0.937 ± 0.007	$+0.003$	$+0.16$
UD/m2	42	L	Laplacian (d12)	PauliXZ (d10)	0.942 ± 0.008	0.940 ± 0.006	-0.002	-0.18
UD/m2	123	S	Laplacian (d12)	PauliXZ (d12)	0.917 ± 0.020	0.926 ± 0.013	$+0.008$	$+0.33$
UD/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.943 ± 0.007	0.940 ± 0.010	-0.004	-0.28
UD/m2	123	L	Laplacian (d12)	Z-map (d12)	0.948 ± 0.003	0.947 ± 0.005	-0.001	-0.20
UD/m2	999	S	Laplacian (d12)	PauliXZ (d12)	0.923 ± 0.023	0.927 ± 0.009	$+0.004$	$+0.16$
UD/m2	999	M	Laplacian (d12)	PauliXZ (d12)	0.935 ± 0.013	0.939 ± 0.007	$+0.004$	$+0.26$
UD/m2	999	L	Laplacian (d12)	Z-map (d12)	0.947 ± 0.004	0.946 ± 0.004	-0.001	-0.22
UD/nat	42	S	Linear (d4)	ZZ-r2 (d10)	0.781 ± 0.040	0.809 ± 0.017	$+0.028$	$+0.65$
UD/nat	42	M	Laplacian (d12)	Z-map (d12)	0.829 ± 0.019	0.838 ± 0.014	$+0.009$	$+0.38$
UD/nat	42	L	Laplacian (d12)	Z-map (d12)	0.829 ± 0.011	0.829 ± 0.008	$+0.000$	$+0.01$
UD/nat	123	S	Laplacian (d12)	ZZ-r2 (d6)	0.775 ± 0.026	0.818 ± 0.018	$+0.043$	$+1.37$
UD/nat	123	M	Laplacian (d12)	ZZ-r1 (d10)	0.790 ± 0.020	0.812 ± 0.007	$+0.022$	$+1.04$
UD/nat	123	L	Laplacian (d12)	ZZ-r1 (d12)	0.809 ± 0.012	0.825 ± 0.011	$+0.016$	$+0.97$
UD/nat	999	S	Linear (d4)	ZZ-r2 (d6)	0.759 ± 0.051	0.800 ± 0.018	$+0.042$	$+0.77$
UD/nat	999	M	Laplacian (d12)	ZZ-r2 (d10)	0.788 ± 0.016	0.809 ± 0.014	$+0.021$	$+0.95$
UD/nat	999	L	Laplacian (d12)	ZZ-r1 (d12)	0.804 ± 0.006	0.826 ± 0.013	$+0.023$	$+1.63$
UR/m2	42	S	Laplacian (d12)	Z-map (d12)	0.909 ± 0.025	0.909 ± 0.015	-0.000	-0.01
UR/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.931 ± 0.019	0.935 ± 0.008	$+0.004$	$+0.18$
UR/m2	42	L	Laplacian (d12)	PauliXZ (d12)	0.944 ± 0.005	0.943 ± 0.006	-0.001	-0.12
UR/m2	123	S	Laplacian (d12)	PauliXZ (d12)	0.946 ± 0.014	0.942 ± 0.007	-0.004	-0.24
UR/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.951 ± 0.011	0.948 ± 0.008	-0.004	-0.26
UR/m2	123	L	Laplacian (d12)	Z-map (d12)	0.954 ± 0.007	0.954 ± 0.006	-0.000	-0.03
UR/m2	999	S	Laplacian (d12)	Z-map (d12)	0.915 ± 0.016	0.918 ± 0.014	$+0.003$	$+0.12$
UR/m2	999	M	Laplacian (d12)	Z-map (d12)	0.947 ± 0.006	0.944 ± 0.008	-0.003	-0.33
UR/m2	999	L	Laplacian (d12)	Z-map (d12)	0.956 ± 0.003	0.948 ± 0.007	-0.008	-1.03
UR/nat	42	S	Linear (d12)	ZZ-r2 (d6)	0.770 ± 0.019	0.778 ± 0.013	$+0.008$	$+0.36$
UR/nat	42	M	Laplacian (d10)	ZZ-r1 (d4)	0.809 ± 0.035	0.815 ± 0.010	$+0.007$	$+0.18$
UR/nat	42	L	Laplacian (d10)	ZZ-r1 (d4)	0.812 ± 0.020	0.815 ± 0.008	$+0.003$	$+0.12$
UR/nat	123	S	Linear (d12)	ZZ-r1 (d6)	0.795 ± 0.012	0.797 ± 0.018	$+0.002$	$+0.11$
UR/nat	123	M	Linear (d10)	ZZ-r1 (d4)	0.800 ± 0.018	0.802 ± 0.024	$+0.003$	$+0.09$
UR/nat	123	L	Laplacian (d12)	ZZ-r1 (d4)	0.818 ± 0.018	0.811 ± 0.009	-0.007	-0.34
UR/nat	999	S	Linear (d12)	ZZ-r1 (d6)	0.780 ± 0.020	0.781 ± 0.008	$+0.000$	$+0.02$
UR/nat	999	M	Linear (d12)	ZZ-r2 (d6)	0.783 ± 0.014	0.809 ± 0.015	$+0.026$	$+1.26$
UR/nat	999	L	Laplacian (d8)	ZZ-r2 (d6)	0.804 ± 0.015	0.814 ± 0.008	$+0.010$	$+0.59$

Table A4 Network flows, GP classifier: full Best-by-OOD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
TS/m2	42	S	Laplacian (d12)	ZZ-r1 (d8)	0.906 ± 0.019	0.954 ± 0.011	+0.049	+2.27
TS/m2	42	M	Laplacian (d8)	ZZ-r1 (d12)	0.921 ± 0.004	0.957 ± 0.005	+0.037	+5.70
TS/m2	42	L	Laplacian (d8)	ZZ-r1 (d8)	0.931 ± 0.007	0.964 ± 0.006	+0.034	+3.58
TS/m2	123	S	Laplacian (d8)	ZZ-r1 (d10)	0.910 ± 0.056	0.958 ± 0.006	+0.048	+0.86
TS/m2	123	M	Laplacian (d4)	ZZ-r1 (d12)	0.932 ± 0.007	0.958 ± 0.007	+0.025	+2.59
TS/m2	123	L	Laplacian (d4)	ZZ-r1 (d10)	0.934 ± 0.004	0.960 ± 0.003	+0.026	+4.84
TS/m2	999	S	Laplacian (d6)	ZZ-r1 (d10)	0.918 ± 0.021	0.954 ± 0.009	+0.036	+1.58
TS/m2	999	M	Laplacian (d4)	ZZ-r1 (d10)	0.931 ± 0.002	0.961 ± 0.006	+0.030	+4.49
TS/m2	999	L	Laplacian (d8)	ZZ-r1 (d10)	0.941 ± 0.003	0.968 ± 0.003	+0.027	+6.65
TS/nat	42	S	Laplacian (d8)	ZZ-r1 (d8)	0.981 ± 0.005	0.990 ± 0.002	+0.009	+1.67
TS/nat	42	M	Laplacian (d8)	ZZ-r1 (d10)	0.987 ± 0.002	0.991 ± 0.003	+0.004	+1.08
TS/nat	42	L	Laplacian (d8)	ZZ-r1 (d12)	0.983 ± 0.003	0.992 ± 0.002	+0.009	+2.30
TS/nat	123	S	Laplacian (d8)	ZZ-r1 (d8)	0.981 ± 0.017	0.992 ± 0.003	+0.011	+0.65
TS/nat	123	M	Laplacian (d8)	ZZ-r1 (d10)	0.985 ± 0.004	0.993 ± 0.002	+0.008	+1.72
TS/nat	123	L	Laplacian (d8)	ZZ-r1 (d12)	0.985 ± 0.004	0.992 ± 0.002	+0.007	+1.64
TS/nat	999	S	Laplacian (d10)	ZZ-r1 (d10)	0.965 ± 0.027	0.980 ± 0.015	+0.015	+0.48
TS/nat	999	M	Laplacian (d8)	ZZ-r1 (d10)	0.984 ± 0.005	0.991 ± 0.003	+0.007	+1.20
TS/nat	999	L	Laplacian (d8)	ZZ-r1 (d10)	0.988 ± 0.002	0.992 ± 0.003	+0.004	+1.41
UD/m2	42	S	Laplacian (d12)	PauliXZ (d12)	0.894 ± 0.028	0.904 ± 0.016	+0.009	+0.29
UD/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.911 ± 0.007	0.934 ± 0.009	+0.023	+1.96
UD/m2	42	L	Laplacian (d12)	Z-map (d12)	0.927 ± 0.008	0.939 ± 0.010	+0.013	+1.02
UD/m2	123	S	Laplacian (d12)	Z-map (d12)	0.889 ± 0.019	0.917 ± 0.019	+0.028	+1.04
UD/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.916 ± 0.016	0.937 ± 0.007	+0.022	+1.24
UD/m2	123	L	Laplacian (d12)	Z-map (d12)	0.932 ± 0.014	0.945 ± 0.007	+0.013	+0.86
UD/m2	999	S	Laplacian (d12)	Z-map (d12)	0.878 ± 0.022	0.911 ± 0.031	+0.033	+0.87
UD/m2	999	M	Laplacian (d12)	PauliXZ (d12)	0.914 ± 0.011	0.931 ± 0.009	+0.017	+1.19
UD/m2	999	L	Laplacian (d12)	PauliXZ (d12)	0.936 ± 0.009	0.944 ± 0.003	+0.008	+0.82
UD/nat	42	S	Laplacian (d12)	ZZ-r2 (d6)	0.764 ± 0.023	0.801 ± 0.023	+0.037	+1.13
UD/nat	42	M	Laplacian (d12)	PauliXZ (d12)	0.813 ± 0.010	0.829 ± 0.008	+0.016	+1.20
UD/nat	42	L	Laplacian (d12)	Z-map (d12)	0.818 ± 0.010	0.831 ± 0.009	+0.013	+0.97
UD/nat	123	S	Laplacian (d12)	ZZ-r2 (d10)	0.775 ± 0.027	0.813 ± 0.017	+0.038	+1.20
UD/nat	123	M	Laplacian (d12)	ZZ-r2 (d6)	0.781 ± 0.020	0.805 ± 0.016	+0.024	+0.93
UD/nat	123	L	Laplacian (d12)	ZZ-r1 (d12)	0.800 ± 0.005	0.816 ± 0.006	+0.016	+2.09
UD/nat	999	S	Linear (d4)	ZZ-r2 (d6)	0.766 ± 0.053	0.797 ± 0.021	+0.032	+0.55
UD/nat	999	M	Laplacian (d12)	ZZ-r2 (d12)	0.784 ± 0.018	0.806 ± 0.013	+0.022	+1.02
UD/nat	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.805 ± 0.006	0.822 ± 0.012	+0.017	+1.31
UR/m2	42	S	Laplacian (d12)	Z-map (d12)	0.880 ± 0.021	0.902 ± 0.022	+0.022	+0.73
UR/m2	42	M	Laplacian (d12)	Z-map (d12)	0.901 ± 0.008	0.925 ± 0.009	+0.024	+1.99
UR/m2	42	L	Laplacian (d12)	PauliXZ (d12)	0.929 ± 0.012	0.942 ± 0.003	+0.013	+1.05
UR/m2	123	S	Laplacian (d12)	Z-map (d12)	0.905 ± 0.012	0.926 ± 0.023	+0.021	+0.81
UR/m2	123	M	Laplacian (d12)	Z-map (d12)	0.914 ± 0.010	0.946 ± 0.010	+0.032	+2.29
UR/m2	123	L	Laplacian (d12)	PauliXZ (d12)	0.946 ± 0.004	0.953 ± 0.006	+0.007	+0.91
UR/m2	999	S	Laplacian (d10)	PauliXZ (d12)	0.883 ± 0.023	0.910 ± 0.025	+0.027	+0.81
UR/m2	999	M	Laplacian (d12)	Z-map (d12)	0.912 ± 0.004	0.941 ± 0.008	+0.029	+3.13
UR/m2	999	L	Laplacian (d10)	PauliXZ (d12)	0.937 ± 0.014	0.949 ± 0.003	+0.012	+0.84
UR/nat	42	S	Poly-2 (d4)	PauliXZ (d6)	0.760 ± 0.020	0.784 ± 0.024	+0.025	+0.79
UR/nat	42	M	Laplacian (d10)	ZZ-r1 (d4)	0.795 ± 0.040	0.817 ± 0.013	+0.022	+0.51
UR/nat	42	L	Laplacian (d8)	ZZ-r1 (d4)	0.796 ± 0.007	0.813 ± 0.005	+0.017	+1.95
UR/nat	123	S	Poly-2 (d12)	ZZ-r2 (d6)	0.780 ± 0.024	0.797 ± 0.016	+0.017	+0.59
UR/nat	123	M	Linear (d12)	ZZ-r1 (d4)	0.794 ± 0.014	0.803 ± 0.020	+0.008	+0.34
UR/nat	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.810 ± 0.023	0.812 ± 0.019	+0.002	+0.06
UR/nat	999	S	Poly-2 (d12)	ZZ-r2 (d6)	0.777 ± 0.016	0.782 ± 0.005	+0.005	+0.30
UR/nat	999	M	Poly-2 (d12)	ZZ-r2 (d6)	0.780 ± 0.017	0.814 ± 0.016	+0.034	+1.51
UR/nat	999	L	Laplacian (d8)	ZZ-r1 (d4)	0.790 ± 0.010	0.807 ± 0.011	+0.017	+1.10

Statements and Declarations.

Funding. The authors received no specific funding for the submitted work.

Competing Interests. The authors declare that they have no competing interests.

Author Contributions. Roberto Fernández-Barrios: Conceptualization, Methodology, Software, Data curation, Formal analysis, Visualization, Investigation, Writing–original draft, Writing–review & editing. Iker Pastor-López: Supervision, Validation, Writing–review & editing. Asier González-Santocildes: Validation, Project administration, Writing–review & editing. Pablo Garcia Bringas: Supervision, Resources, Writing–review & editing.

Data Availability. This study is based on the public EMBER Anderson and Roth (2018), UNSW-NB15 Moustafa and Slay (2015), and ToN-IoT Alsaedi et al. (2020) benchmark datasets. The derived split definitions, exporters, shift constructions, aggregated results, experiment manifests, and summary tables used in this work are publicly available in the archived project repository: https://github.com/roberto-fernandez-barrios/kernel_shift_framework. Release DOI: https://doi.org/10.5281/zenodo.19152497.

Code Availability. The code used to generate the shift splits, create q-splits, run the classical and quantum kernel experiments, aggregate repeated-run results, and reproduce the reported tables and figures is publicly available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework. Archived release DOI: https://doi.org/10.5281/zenodo.19152497.

Ethics Approval. Not applicable.

Consent to Participate. Not applicable.

Consent for Publication. Not applicable.

References

Alsaedi, A., Moustafa, N., Tari, Z., Mahmood, A., Anwar, A.: TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems. IEEE Access **8**, 165130–165150 (2020) https://doi.org/10.1109/ACCESS.2020.3022862

Anderson, H.S., Roth, P.: Ember: An open dataset for training static PE malware machine learning models. arXiv **abs/1804.04637** (2018)

Bowles, J., Ahmed, S., Schuld, M.: Better than classical? the subtle art of benchmarking quantum machine learning models. arXiv **abs/2403.07059** (2024)

Barbero, F., Pendlebury, F., Pierazzi, F., Cavallaro, L.: Transcending TRANSCEND: Revisiting malware classification in the presence of concept drift. In: 2022 IEEE Symposium on Security and Privacy (SP), pp. 805–823 (2022). https://doi.org/10.1109/SP46214.2022.9833659

Caro, M.C., Huang, H.-Y., Cerezo, M., Sharma, K., Sornborger, A., Cincio, L., Coles, P.J.: Generalization in quantum machine learning from few training data. Nature Communications **13**, 4919 (2022) https://doi.org/10.1038/s41467-022-32550-3

Cortes, C., Mohri, M., Rostamizadeh, A.: Algorithms for learning kernels based on centered alignment. Journal of Machine Learning Research **13**(28), 795–828 (2012)

Canatar, A., Peters, E., Pehlevan, C., Wild, S.M., Shaydulin, R.: Bandwidth enables generalization in quantum kernel models. Transactions on Machine Learning Research (2023)

Cristianini, N., Shawe-Taylor, J., Elisseeff, A., Kandola, J.: On kernel-target alignment. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 14

(2001)

Cortes, C., Vapnik, V.: Support-vector networks. Machine Learning **20**(3), 273–297 (1995) https://doi.org/10.1007/BF00994018

Cerezo, M., Verdon, G., Huang, H.-Y., Cincio, L., Coles, P.J.: Challenges and opportunities in quantum machine learning. Nature Computational Science **2**, 567–576 (2022) https://doi.org/10.1038/s43588-022-00311-3

Gil-Fuster, E., Eisert, J., Bravo-Prieto, C.: Understanding quantum machine learning also requires rethinking generalization. Nature Communications **15**, 2277 (2024) https://doi.org/10.1038/s41467-024-45882-z

Gulrajani, I., Lopez-Paz, D.: In search of lost domain generalization. In: International Conference on Learning Representations (2021). https://openreview.net/forum?id=lQdXeXDoWtI

Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H., McClean, J.R.: Power of data in quantum machine learning. Nature Communications **12**, 2631 (2021) https://doi.org/10.1038/s41467-021-22539-9

Havlíček, V., Córcoles, A.D., Temme, K., Harrow, A.W., Kandala, A., Chow, J.M., Gambetta, J.M.: Supervised learning with quantum-enhanced feature spaces. Nature **567**, 209–212 (2019) https://doi.org/10.1038/s41586-019-0980-2

Harang, R., Rudd, E.M.: Sorel-20m: A large scale benchmark dataset for malicious PE detection. arXiv **abs/2012.07634** (2020)

Joyce, R.J., Miller, G., Roth, P., Zak, R., Zaresky-Williams, E., Anderson, H., Raff, E., Holt, J.: Ember2024 – a benchmark dataset for holistic evaluation of malware classifiers. arXiv **abs/2506.05074** (2025)

Koh, P.W., Sagawa, S., Marklund, H., Xie, S.M., Zhang, M., Balsubramani, A., Hu, W., Yasunaga, M., Phillips, R.L., Gao, I., Lee, T., David, E., Stavness, I., Guo, W., Earnshaw, B., Haque, I., Beery, S.M., Leskovec, J., Kundaje, A., Pierson, E., Levine, S., Finn, C., Liang, P.: Wilds: A benchmark of in-the-wild distribution shifts. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 5637–5664. PMLR, ??? (2021). https://proceedings.mlr.press/v139/koh21a.html

Li, A.S., Iyengar, A., Kundu, A., Bertino, E.: Revisiting concept drift in windows malware detection: Adaptation to real drifted malware with minimal samples. In: Network and Distributed System Security Symposium (NDSS) (2025). https://doi.org/10.14722/ndss.2025.240830

Lazaro, J., Vazquez, J.-I., Garcia-Bringas, P.: Dissecting quantum reinforcement learning: A systematic evaluation of key components. arXiv **abs/2511.17112** (2025) https://doi.org/10.48550/arXiv.2511.17112

Moustafa, N., Slay, J.: UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set),. In: 2015 Military Communications and Information Systems Conference (MilCIS), pp. 1–6 (2015). https://doi.org/10.1109/MilCIS.2015.7348942

Quiñonero-Candela, J., Sugiyama, M., Schwaighofer, A., Lawrence, N.D. (eds.): Dataset Shift in Machine Learning. MIT Press, Cambridge, MA (2008). https://doi.org/10.7551/mitpress/9780262170055.001.0001

Roy, O., Vetterli, M.: The effective rank: A measure of effective dimensionality. In: 2007 15th European Signal Processing Conference (EUSIPCO), pp. 606–610 (2007)

Rasmussen, C.E., Williams, C.K.I.: Gaussian Processes for Machine Learning. MIT

Press, Cambridge, MA (2006). https://doi.org/10.7551/mitpress/3206.001.0001

Schuld, M.: Supervised quantum machine learning models are kernel methods. arXiv **abs/2101.11020** (2021)

Shimodaira, H.: Improving predictive inference under covariate shift by weighting the log-likelihood function. Journal of Statistical Planning and Inference **90**(2), 227–244 (2000) https://doi.org/10.1016/S0378-3758(00)00115-4

Schuld, M., Killoran, N.: Quantum machine learning in feature hilbert spaces. Physical Review Letters **122**(4), 040504 (2019) https://doi.org/10.1103/PhysRevLett.122.040504

Sugiyama, M., Krauledat, M., Müller, K.-R.: Covariate shift adaptation by importance weighted cross validation. Journal of Machine Learning Research **8**(35), 985–1005 (2007)

Schnabel, J., Roth, M.: Quantum kernel methods under scrutiny: A benchmarking study. Quantum Machine Intelligence **7**, 58 (2025) https://doi.org/10.1007/s42484-025-00273-5

Schölkopf, B., Smola, A.J.: Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond. MIT Press, Cambridge, MA (2002)

Schuld, M., Sweke, R., Meyer, J.J.: Effect of data encoding on the expressive power of variational quantum-machine-learning models. Physical Review A **103**(3), 032430 (2021) https://doi.org/10.1103/PhysRevA.103.032430

Shaydulin, R., Wild, S.M.: Importance of kernel bandwidth in quantum machine learning. Physical Review A **106**(4), 042407 (2022) https://doi.org/10.1103/PhysRevA.106.042407

Yao, H., Choi, C., Cao, B., Lee, Y., Koh, P.W., Finn, C.: Wild-time: A benchmark of in-the-wild distribution shift over time. In: Advances in Neural Information Processing Systems (2022). Datasets and Benchmarks Track

Yang, L., Ciptadi, A., Laziuk, I., Ahmadzadeh, A., Wang, G.: Bodmas: An open dataset for learning based temporal analysis of PE malware. In: 2021 IEEE Security and Privacy Workshops (SPW), pp. 78–84 (2021). https://doi.org/10.1109/SPW53761.2021.00020

Zhou, K., Liu, Z., Qiao, Y., Xiang, T., Loy, C.C.: Domain generalization: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **45**(4), 4396–4415 (2023) https://doi.org/10.1109/TPAMI.2022.3195549